Introduction to projects

ETSN20 Software Testing

http://cs.lth.se/etsn20

Introduction by Per Runeson

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Project learning objectives

- Learn a specific area of software testing
- Collect and summarize research information
- Critical thinking beyond the written information
- Present information in a structured way



Project

- **Practical**: Use an existing technique or testing tool. Focus on technique/tool evaluation or testing of software system.
- **Research**: solve a research problem; survey the state-of-the-art and identify the research problems in some area; develop and justify an extension to an existing technique; etc.



9 Practical Project Areas 5
9.1 Evaluate tools for software testing or review 5
9.2 Evaluate test methods 5
9.3 Develop an automated test suite 5
10 Research Project Areas 6
10.1 How is testing performed for Agile processes? 6
10.2 Can test cases be generated automatically from the code? 6
10.3 How to use LLMs to generate (good) tests? 6
10.4 How to test machine learning systems? 7
10.5 How to test interpretability in machine learning system? 7
10.6 What are the minimal test practices needed in a start-up company? 8
10.7 Is it more efficient to generate test cases from a model? 8
10.8 How do you test for security? 8
10.9 Do static code analysis tools help improve software code quality? 8
10.10 How do you know that your test cases are effective? 9
10.11 Flaky or non-deterministic tests 9
10.12 How to test the energy consumption of software systems? 9
10.13 How organize assertions in test cases? 10



Project activities

- Decide project type and subject
- Establish a collaboration agreement
- Find literature
- Outline the report [-> supervisor]
- Read; perform evaluation and practical work
- Write the report [in English!]
- Consult the supervisor
- Present the report



Find literature

- **Different fora has different credibility**
- Scientific fora: *rigorously reviewed*
  - Journals
  - Conferences
  - Workshops
- Non-scientific fora:
  - Textbooks
  - Journalistist material
  - White papers
  - Web pages (inkl Wikipedia!)
  - …

- **To search for scientific papers**
  - http:/www.lub.lu.se
  - Detective work:
    - Search broad
    - Select
    - Search deep

    ➢ Iterate

    - Key words
    - Authors
    - References
    - Fora
- **GenAI – use with care**



Build on other's work

Systematic Literature Reviews (SLR)

- Systematically collected and summarized research on a **specific research question**

Mapping study (MS)

- Collected and summarized research on **a broader topic**

[Garousi and Mäntylä, 2016, Zhang et al. 2022, Wang et al., 2024]



Consider Threats to Validity when Reading

- Everything you read is not true – be critical!
- ”Threats to Validity” sections should openly and honestly report and discuss limitations, reliability, generalizability (wider applicability), etc.
- If you use non-peer reviewed material (blogs, white papers, GenAI summaries etc.) you have to assess the quality on your own!



Report

- 5-7 pages in IEEE format (see Project description for details)

- Abstract
- Introduction
- Description of the chosen area
- Analysis
- Conclusion
- Contribution statement
- References



Resources on Academic Writing

https://canvas.education.lu.se/courses/12628
(mostly in Swedish)

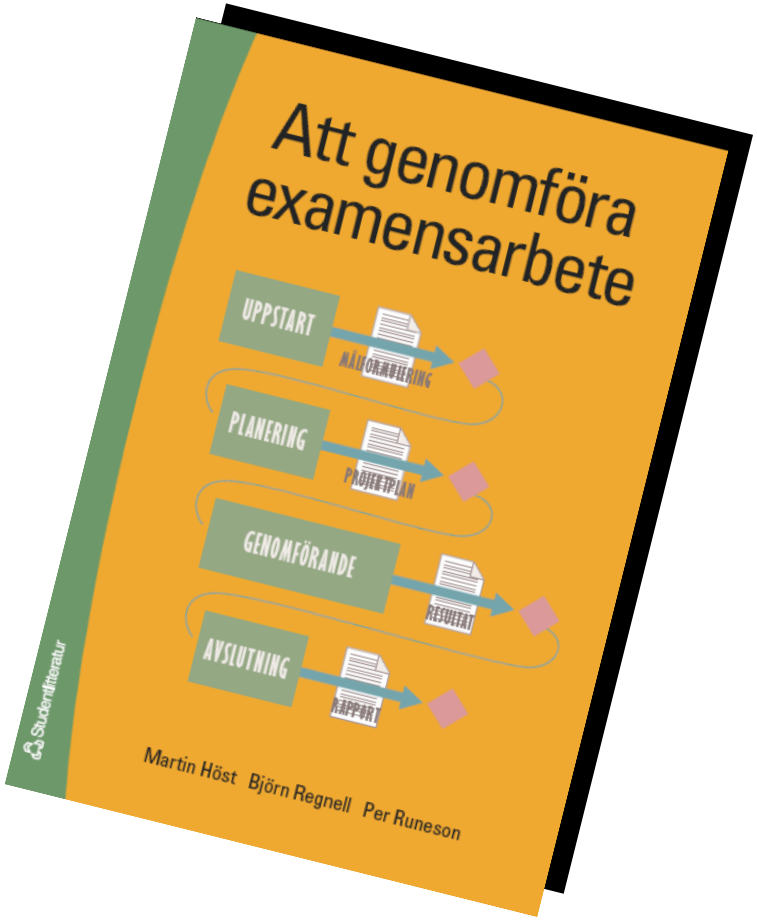
https://awelu.srv.lu.se/ (English)

MOOC on Academic Writing

https://youtube.com/playlist?list=
PLeVxAnFsasIqIc8b03kHA3tw-xfIwgO2M

Useful tips

1. Inledning
2. Start och målformulering
3. Att välja metodik
4. Planering och uppföljning
5. Litteraturstudier
6. Verktyg för genomförande
7. Rapporten
8. Muntlig presentation
9. Opposition



https://github.com/lunduniversity/thesis-guide/releases/download/v1.0/thesis-guide-v1.pdf

Cooperate, don’t copy

- Submit report in Canvas – control via Inspera
- **GenAI**?
  - **You may** use GenAI as a sparring partner. This means that GenAI can be a useful tool during the beginning stages of your research.
  - **You may not** let GenAI write your text, or to let it rewrite an already existing text.
  - **What do want to learn?** It is unlikely that letting a chatbot do your work will improve the skills you’re trying to gain.

https://www.staff.lu.se/sites/staff.lu.se/files/2026-08/guidelines-for-the-use-of-generative-ai-in-first-and-second-cycle-education-at-lund-university.pdf



Limitations and risks of AI use

- Reduced learning through shortcuts.
- Errors and bias.
- Insufficient source citation.
- Unnecessary extra work – iterations, prompt engineering

Raw AI text is often less precise and of lower quality than text you formulate yourself.



Guidelines for AI use

- If AI is used, it is important to:
  - The use is transparent
  - AI must not replace your own reasoning
  - You must not mislead reviewers and examiners
  - Critical review required



Assessment

- Project graded (U, G, VG) by weighting
  - 40% work (scope, depth etc.)
  - 40% content (structure, flow, etc.)
  - 0% form (template, ref style, language etc.)
  - 20% presentation (clarity, timeliness etc.)
- Only G if second round needed



Project – important dates

- Decide on a subject: Monday Nov 9 (noon!)
- Outline: Tuesday Nov 17
- Consult the supervisor course w 4-5 (book time by Nov 27!)
- Report: Friday Dec 11
- Presentation: Thursday Dec 17, 13-17
- Update report – if needed: Jan 8



Project selection procedure

1. Move to the table, labeled with your favorite project topic
2. If you are 3-4 students, register in Canvas for the project
3. If you are too many or too few, discuss with me

4. Start establishing a collaboration agreement



Collaboration agreement

a) ambition level within the group (pass/pass with distinction
b) communication format and working platform
c) meeting and work schedule
d) task breakdown and division

Submit in Canvas in pdf format

